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# Introduction

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Course of Cybersecurity, 2025/2026

# Security

- Making a system (a software, a computer, a network, etc.) secure, requires three types of security:
  - **physical security**
    - physically limit the access to the system
      - servers behind a locked door and only a privileged set of employees have access to it
      - use of cameras, card readers, and biometric locks
    - protection again information leakage
      - e.g. by shredding documents before they're thrown away
  - **technological security**
    - communication and data security
    - software security
      - application security
      - OS security
  - **good policies and practices**

# Security Service

- It is something that enhances the security of a system and the information transfer
  - **aims to protect data, systems, user information**
  - **intended to counter security attacks**
- A processing or communication service that is provided by a system or other entities to give a specific kind of protection to system resources
  - **RFC 4949 , "Internet Security Glossary"**
    - https://tools.ietf.org/html/rfc4949
- Makes use of one or more security mechanisms to provide the service
- Replicates functions normally associated with physical objects/documents
  - **e.g. signatures, dates, proof of reception, notarization, recording, etc.**

# Security Services (cont.)

- Some security services:
  - **Confidentiality**
  - **Data integrity and message authentication (or authenticity)**
  - **Peer entity authentication (or identification)**
  - **Authorization and access control**
  - **System integrity and availability**
  - **Accountability and non-repudiation**
  - **Anonymity**

# Security Services (cont.)

- Confidentiality
  - **protects data against unauthorized disclosure**
    - It is the property that information is not made available to unauthorized entities
    - related to
      - data
        - » data confidentiality
      - entities involved in the communication
        - » anonymity

# Security Services (cont.)

- Data integrity and message authentication
  - **data integrity**
    - the property that data has not been changed, destroyed, or lost in an unauthorized or accidental manner
      - protects against unauthorized changes to data by ensuring that changes to data are detectable
      - in general it can only detect a change
  - **data origin authentication**
    - provides for the corroboration of the source of a data unit
      - this service verifies the identity of a system entity that is claimed to be the original source of received data
    - usually provided together with data integrity → message authentication
  - **message authentication (authenticity)**
    - both data integrity and origin authentication
    - in general, authenticity implies integrity but integrity doesn't imply authenticity

# Security Services (cont.)

- Identification (peer entity authentication)
  - **verification of the identity of a peer entity**
    - before the establishment of a communication or the access to a resource/service

- Authorization and access control
  - **authorization**
    - verification of the permission to access a resource or system
      - manage access rights/privileges
  - **access control**
    - ability to limit and control the access to a systems
      - protection of system resources against unauthorized access

# Security Services (cont.)

- System integrity and availability
  - **system integrity**
    - the quality that a system has when it can perform its intended function
      - protects system resources against unauthorized change, loss, or destruction
  - **availability**
    - protects a system to ensure its availability
      - addresses the security concerns raised by denial-of-service (DoS) attacks

# Security Services (cont.)

- Accountability and non-repudiation
  - **Accountability and audit**
    - Accountability refers to the ability to trace actions to specific users or entities
    - audit refers to the process of reviewing and analyzing logs, records, or actions logged by the accountability
  - **Non-repudiation**
    - provides protection against false denial of an action
      - it provides evidence that can be stored and later presented to a third party
    - in case of a communication, it prevents either sender or receiver from denying a transmitted message
      - the receiver can prove that the sender in fact sent the message
        - » non-repudiation with proof of origin
      - the sender can prove that the receiver in fact received the message
        - » non-repudiation with proof of receipt

# Security Mechanisms

- Security services are provided by means of different security functions/ mechanisms
  - **they can be included in appropriate communication layer**

- Examples of security mechanisms are:
  - **authentication exchange**
  - **data integrity check**
  - **enciphering**
  - **digital signature**
  - **notarization (third-party authentication)**
  - **access control**
  - **routing control**
  - **traffic padding**
  - **etc.**